Design of p53-Targeting Peptides Using Language Model-Based Sequence Generation

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Abstract

The tumor suppressor p53, frequently mutated in human cancers, represents a critical yet challenging therapeutic target due to its intrinsically disordered nature. Here, a novel approach to designing peptide binders for p53 using PepMLM, a sequence-based deep learning algorithm, is presented. Through computational screening of 200 candidate sequences, a 12-amino acid peptide (SDEPLLGGMPRT) was identified that was predicted to bind p53 with relatively high affinity. The peptide was expressed as a SUMO-tagged fusion protein in E. coli, purified via nickel affinity chromatography, and biotinylated for binding studies. ELISA analysis revealed that the designed peptide binds to p53 with a dissociation constant (K_d) of 0.39 nM, stronger than the PBS control (K_d = 0.62 nM); however, further statistical analysis reveals no statistically significant binding. The findings highlight the potential for a novel sequence-driven approach for developing peptide binders against proteins lacking stable structural domains, expanding the repertoire of targetable disease-related proteins. The approach circumvents the limitations of structure-based design methods and establishes a promising pipeline for developing peptide-based therapeutics. These findings not only advance the ability to target p53 but also suggest broader applications for designing peptide binders against other challenging therapeutic targets in cancer and other diseases.

Introduction

The tumor suppressor p53, encoded by the TP53 gene, stands as one of the most critical and extensively researched proteins in cancer biology. Widely known as the "guardian of the genome," p53 conducts cellular responses to various stresses by regulating genes essential for cell cycle control, DNA repair, and programmed cell death. Because the TP53 gene is known to be the most frequently mutated gene in human cancers, the fundamental role of p53 in preventing cancer development has been underscored by extensive research showing that its function is compromised either through direct mutations or indirect pathway disruption[1]. Additionally, recent studies have shown that 80% of these mutations are missense mutations occurring within the central sequence-specific DNA binding domain clustered around only a few amino acid hotspots[2]. As such, binders to p53 have immense therapeutic potential. For instance, an effective binder can be attached to structures like E3 ubiquitin ligases to degrade diseased proteins.[19]

While this prevalence makes p53 an attractive therapeutic target, its complex regulatory network and the diverse mechanisms of its inactivation have challenged efforts to develop effective treatments. In addition, p53's structural characteristics make the therapeutic targeting of p53 difficult. As an intrinsically disordered protein, p53 lacks well-defined binding pockets typically exploited for drug development, making traditional small-molecule approaches particularly complicated[3]. Adding to this complexity is the protein's regulation by the ubiquitin-proteasomal system, specifically through MDM2-mediated ubiquitination, which maintains strong control over p53 levels in cells[3]. Currently, while researchers have developed various MDM2 inhibitors and peptide-based approaches to stabilize p53, these interventions typically affect only one aspect of p53's complex regulatory network, resulting in limited therapeutic success. Recent identification of p53-binding peptides has shown promise, but concerns about specificity and potential off-target effects have hindered their clinical development.

The emergence of sophisticated protein design tools has revolutionized the ability to create targeted therapeutic molecules. Advanced computational platforms, including deep learning algorithms such as PepMLM[4] and structure-based design methods such as RFDiffusion, now enable the rational design of peptide sequences with precise binding properties that have high specificity and affinities[5]. These tools have opened new possibilities for targeting previously "undruggable"[6] proteins through the development of selective peptide-based inhibitors. Building on these technological advances, this study focuses on designing a compact 12-amino acid peptide that specifically targets and stabilizes p53 in its active conformation binding with nanomolar affinity and blocking key degradation pathways. This approach represents a novel strategy for therapeutic p53 modulation, potentially offering a more selective and effective treatment option for cancers with compromised p53 function.

Results

Computational Design and Screening:

To design a *de novo* candidate binder, pepMLM was used. 200 candidates were generated and ranked according to the lowest pseudo-perplexity score. The three best-scoring binders are shown here with their respective perplexities.

Table 1. **Binder sequences and their corresponding pseudo-perplexity values**. Highlights differences in model-predicted sequence confidence.

Binder Sequence	Pseudo-Perplexity
SDEPLLGGMPRT	7.70
SDTPLYSGCPRT	8.54
GDYYPLSGCLGT	8.90

The most promising binder, highlighted in Table 1, was selected for further binder affinity testing. The sequence was co-folded with p53 protein utilizing AlphaFold3 to deliver preliminary binding results. An ipTM score of 0.22 was obtained. However, while this low score may signal a lack of stability in the binding interaction or AlphaFold's lack of confidence in the prediction, it is also likely due to the intrinsically disordered nature of the generated binder.

In vitro peptide screening was planned but not performed due to time constraints. A phage display or yeast display would have screened for multiple candidate binders *in vitro*, using the pool of binders created computationally as a starting point. Because no *in vitro* screening was performed, the top-performing pepMLM binder to proceed with binding affinity analysis. This binder was expressed and purified as described in the methods section.

Binding Affinity Analysis:

To test binding *in vitro*, an ELISA assay was utilized. Two plate setups were employed, one with the target p53 protein and the other with PBS as a negative control. The binding results of the ELISA are shown in Figure 3A.

Observations

ELISA

The expected color change to blue[7] was observed following the addition of TMB-ELISA, and the subsequent addition of H_2SO_4 resulted in the anticipated color change to yellow. Plate saturation trends were generally directionally accurate, with higher concentrations appearing more saturated; however, deviations were noted. Specifically, the stock solution exhibited a lower saturation compared to the first 1:4 dilution.

While the binding affinity of the binder to the p53 protein (0.39 nM), appears to be slightly stronger than that of the binder to the PBS control (0.62 nM), the binding does not appear to be selective as the PBS negative control has high signal relative to the binder-p53 curve. This indicates high levels of nonspecific binding to the plate as well as ineffective binding by the candidate peptide.

Comparisons to Control Groups:

Since only one ELISA test was run for the control plate and the plate containing p53, a statistical test with K_D would be infeasible with one sample. Instead, a paired t-test was run across each concentration of binder, comparing the OD_{450} of the p53 group with that of the PBS control. Pairing was chosen to avoid biasing the statistical test towards higher concentration binder solutions (as lower ones would effectively be negligible in the average in an unpaired t-test). Since the OD_{450} values for the p53 is expected to be higher, a one-sided t-test was used. The null hypothesis was that there was no difference in the mean difference between the control and p53 groups. The alternative hypothesis was that the mean difference between p53 and the control is greater than 0. A t-stat value of -0.18 was obtained with a t-critical value of 2.31. This yielded a p-value of 0.43, which implies that the difference in OD_{450} was not at all statistically significant at any reasonable significance level. Interestingly, the difference in OD_{450} implied that the control group had stronger binding than

the p53 group, while the curve fitting for K_D signaled the opposite. This is likely due to the high variability in the data, which meant the logistic curve fitting did not necessarily represent the data accurately.

PBS was used as a negative control for the binding assay, allowing for nonselective binding to be assessed. In several concentration ranges, namely < 10^1 nM and > 10^3 nM, protein binding (OD_{450}) was higher for PBS than for p53. Additionally, even in the concentration range of higher p53 binding, significant levels of PBS binding were present, suggesting nonselective binding.

Bovine serum albumin (BSA) was intended to be a negative control for the candidate peptide's binding. Typically, BSA acts as a blocking buffer to prevent nonspecific binding[14]. However, at high concentrations, BSA itself can nonspecifically bind to the plate[18]. This was observed in the experimental assay, in which the BSA row experienced the greatest binding, much more so than the candidate binder. To mitigate this in future experiments, a negative control closer in size and properties to the protein of interest should be used.

Discussion

Computational methods have advanced rapidly in recent years, revolutionizing fields such as protein folding[15], analyzing interactive molecular effects[16], and generating *de novo* binders[8, 17]. The integration of *in silico* tools like RF Diffusion into peptide design pipelines has demonstrated remarkable success in generating peptide binders for a variety of disease-associated proteins[9]. The high-throughput nature of these computational approaches enables researchers to identify promising candidate binders with precision before advancing to rigorous *in vitro* evaluations. However, structure-based deep learning models have significant limitations in addressing intrinsically disordered proteins (IDPs), often regarded as "undruggable" due to their lack of stable tertiary structure and the associated challenges in designing effective binders[10].

The results here present a potential advancement in peptide design pipelines, specifically addressing the challenge of targeting IDPs. By employing a sequence-based deep learning model, pepMLM, candidate peptide binders were successfully generated, overcoming the limitations of structure-based models. As a proof of concept, this pipeline was applied to p53, a critical IDP involved in cancer regulation. These results underscore the potential efficacy of sequence-based approaches in expanding the druggable protein landscape. Additional validation steps, including comprehensive *in vitro* and *in vivo* testing, will be essential to confirm the clinical relevance of these binders and their functional impact on p53 activity. This validation could involve the structural characterization of peptide-protein interactions, assessing binding specificity, and evaluating downstream cellular effects.

The findings highlight the promise of integrating sequence-based algorithms into drug discovery pipelines, offering new opportunities to target previously inaccessible proteins. This work represents an important step toward the development of innovative therapeutics for IDPs, underscoring the transformative potential of computational methods in addressing unmet medical needs.

Materials and Methods

Binder Design

Novel p53-binding peptides were generated using the PepMLM algorithm (https://huggingface.co/ChatterjeeLab/PepMLM-650M)[11] by providing the p53 protein sequence as input. Approximately 200 candidate 12-mer peptide sequences were generated and ranked based on their pseudo-perplexity scores, where lower values indicate higher predicted binding affinity. The top-scoring peptide sequence SDEPLLGGMPRT was selected for further experimental validation based on having the lowest perplexity value among all generated candidates. All candidate peptide sequences and their corresponding perplexity scores are provided in Supplementary Table 1.

Plasmid Generation

The p53-binding peptide construct was generated using the pET-24a(+) bacterial expression vector. The vector backbone was digested with NdeI and BamHI restriction enzymes. The peptide sequence (SDEPLLGGMPRT) was designed as part of a larger insert containing the following elements in order: NdeI overhang, start codon, GGGSG linker, SUMO tag, two GGGSG linkers, the peptide binder sequence, two stop codons, and BamHI overhang. This insert sequence was codon-optimized for E. coli expression and ordered as complementary DNA oligonucleotides. The insert was cloned into the digested pET-24a(+) vector using Gibson assembly with the corresponding overhangs. The assembled construct was transformed into NEB Turbo Competent Escherichia coli cells, and positive transformants were selected on LB agar plates containing appropriate antibiotics. Individual colonies were sequence-verified to confirm the correct assembly of the final construct (team3_p53_pMLM_1).

Protein Expression and Purification

The sequence-verified construct was transformed into E. coli BL21(DE3) cells and grown overnight on LB agar plates containing appropriate antibiotics. A single colony was picked to inoculate a 10 mL starter culture, which was grown overnight at 37°C with shaking. The starter culture was used to inoculate a 50 mL expression culture that was grown at 37°C with shaking until reaching an OD_{600} of 0.4-0.6, at which point protein expression was induced with 100 mM IPTG. The induced culture was grown for approximately 16 hours at 37°C.

Cells were harvested by centrifugation at 5000 rpm for 30-45 minutes at 4°C. The cell pellet was resuspended in 8 mL His binding buffer and protein concentration were determined by measuring absorbance at 280 nm (A280 = 19.07 mg/mL). For purification, 2.1 mL of the resuspended solution (containing approximately 40 mg protein) was mixed with an equal volume of Equilibration Wash Buffer. Protein purification was performed using HisPur Ni-NTA Spin Columns (Thermo Scientific). The columns were first equilibrated with 2 mL Equilibration Wash Buffer and centrifuged at 700 × g for 2 minutes at 4°C. The protein sample was then applied to the column and incubated with gentle agitation at 4°C for 30 minutes to allow binding of the His-tagged protein. The column was washed three times with 2 mL Equilibration Wash Buffer, and the purified protein was eluted with 1 mL Elution Buffer containing imidazole. Protein purity was assessed by SDS-PAGE, and concentration was determined by A280 measurements.

Binder Biotinylation

The purified SUMO-tagged peptide binder (MW 14.8 kDa) at a concentration of 1.957 mg/mL was biotinylated using Sulfo-NHS-LC-Biotin. The biotin reagent was equilibrated to room temperature before use. A 10 mM biotin reagent solution was freshly prepared by dissolving 2.0 mg of Sulfo-NHS-LC-Biotin in 360 µL UltraPure distilled water. The amount of biotin reagent needed was calculated for a 20-fold molar excess relative to the protein concentration using the formula:

$$1\ mL\ Binder\ \cdot \frac{1.957\ mg\ Binder}{1\ mL\ Binder} \cdot \frac{1\ mmol\ Binder}{14{,}800\ Binder} \cdot \frac{20\ mmol\ Biotin}{1\ mmol\ Binder} = 0.00264\ mmol\ Biotin$$

Subsequently, the calculated volume of 10 mM biotin reagent was calculated with the following formula:

$$0.00264\ mmol\ Biotin\ \cdot \frac{1\ L\ Biotin}{10\ mmol\ Biotin} \cdot \frac{10^6\ uL\ Biotin}{1\ L\ Biotin} = 264\ uL\ Biotin$$

The calculated volume of 10 mM biotin reagent solution was added to the protein solution and incubated on ice for two hours. Excess biotin was removed using a Zeba desalting column (Thermo, 89882). The column was prepared by removing the storage solution via centrifugation at 1000 × g for 2 minutes, followed by three

washes with 1 mL 1X TBST buffer. The biotinylation reaction mixture was loaded onto the prepared column and centrifuged at 1000 × g for 2 minutes. The final concentration of the biotinylated protein was determined by Nanodrop measurement to be 5.04 mg/mL, assuming approximately equal parts of biotin and protein in the final solution.

ELISA Binding Assay

The binding affinity between p53 and the biotinylated peptide binder was assessed using an enzyme-linked immunosorbent assay (ELISA). Clear polystyrene 96-well plates (Corning) were coated with 50 µL/well of purified p53 protein (5 µg/mL in PBS, pH 7.4) and incubated overnight at 4°C. A control plate was prepared in parallel using PBS only. After coating, plates were washed three times with PBS-T (PBS containing 0.1% Tween-20), followed by blocking with 200 µL/well of 5% nonfat dry milk in PBS overnight at 4°C.

The biotinylated peptide binder (initial concentration 170.27 µM) was serially diluted in PBS to generate eight concentrations ranging from 42.57 µM to 0.0026 µM using 1:4 dilution steps. Each dilution was prepared in 500 µL total volume by combining 125 µL of the previous concentration with 375 µL PBS. After washing the blocked plates three times with PBS-T, 80 µL of each diluted binder solution was added to designated wells in triplicate and incubated at 37°C for 1 hour.

Following binder incubation and three PBS-T washes, HRP-conjugated streptavidin (ThermoFisher, Cat #N100) was diluted 1:20,000 in PBS, and 200 µL was added to each well. Plates were incubated at room temperature with gentle shaking for 30 minutes. After washing, 100 µL of TMB substrate (1-Step Ultra TMB-ELISA) was added to each well, and the plates were incubated in the dark for 10 minutes. The reaction was stopped by adding 100 µL of H_2SO_4, and absorbance was immediately measured at 450 nm using a plate reader. Binding curves and dissociation constants (Kd) were determined using custom Python scripts (available at [Final_ELISA_Kd_Analysis.ipynb]).[13]

Computation Tools and Metrics

To identify potential peptide binders for the tumor suppressor protein p53, PepMLM, a target sequence-conditioned generator of de novo linear peptide binders, was employed. PepMLM utilizes a novel span masking strategy, positioning the entire peptide binder sequence at the C-terminus of the target protein sequence [3]. This approach fine-tunes the ESM-2 protein language model to reconstruct the binder region, achieving low perplexities that match or surpass validated peptide-protein sequence pairs.

The amino acid sequence of human p53 will be retrieved from the UniProt database (UniProt ID: P04637). This sequence will serve as the input for PepMLM, with a designated masking region at the C-terminus to facilitate the generation of potential peptide binders.

PepMLM will generate a diverse set of 200 candidate peptide sequences designed to bind to p53. Each generated peptide will be evaluated based on its perplexity score, a measure of the model's confidence in the sequence's plausibility and potential binding affinity. The top 2-3 candidate peptides exhibiting the lowest perplexity scores were selected for further analysis.

Predicted structure of the P53 :: Binder interaction was performed on the AlphaFold3 server (https://alphafoldserver.com/). Shading corresponds to AlphaFold3's standard confidence metric, plDDT, with low plDDTs (orange) corresponding to low confidence and high plDDTs (Blue) corresponding to high confidence. More specifically, (plDDT < 50) = Orange, (70 > plDDT > 50) = Yellow, (70 > plDDT > 90) = Light Blue, (plDDT > 90) = Blue.

Statistical Analyses

Sample sizes were not predetermined based on statistical methods but were chosen according to the standards of the field (at least three independent biological replicates for each condition), which gave sufficient statistics for the effect sizes of interest. Statistical analysis was performed using the one-tailed paired Student's t-test using Google Sheets. Graphs were generated with standard error bars utilizing the matplotlib library in Python 3. No data were excluded from the analyses. The experiments were not randomized. The investigators were not blinded to allocation during experiments and outcome assessment.

Author Contributions

L.H., T.W., and S.V., built constructs, conducted transfections, and collected binders. B.L., I.M., purified and biotinylated binders. I.M., A.I. prepared the ELISA, while A.I., S.S., performed the ELISA binding assay. B.L., S.S., analyzed ELISA results and performed statistical analyses. I.M., B.L., prepared the figure set. I.M., B.L, A.I., S.S., finalized results and conclusions. All authors wrote and revised the manuscript. P.C. conceived, designed, directed, and supervised the study.

Declarations

The authors have nothing to declare.

Competing Interests Statement

The authors have no competing interests.

Data Availability and Code Availability

All data needed to evaluate the conclusions in the paper are present in the paper and supplementary information. All statistical analysis calculations are contained in this spreadsheet. The code used to generate Figure 3b is given in this Colab notebook. The code used to generate Figure 3a is provided in this Colab notebook.

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Figures:

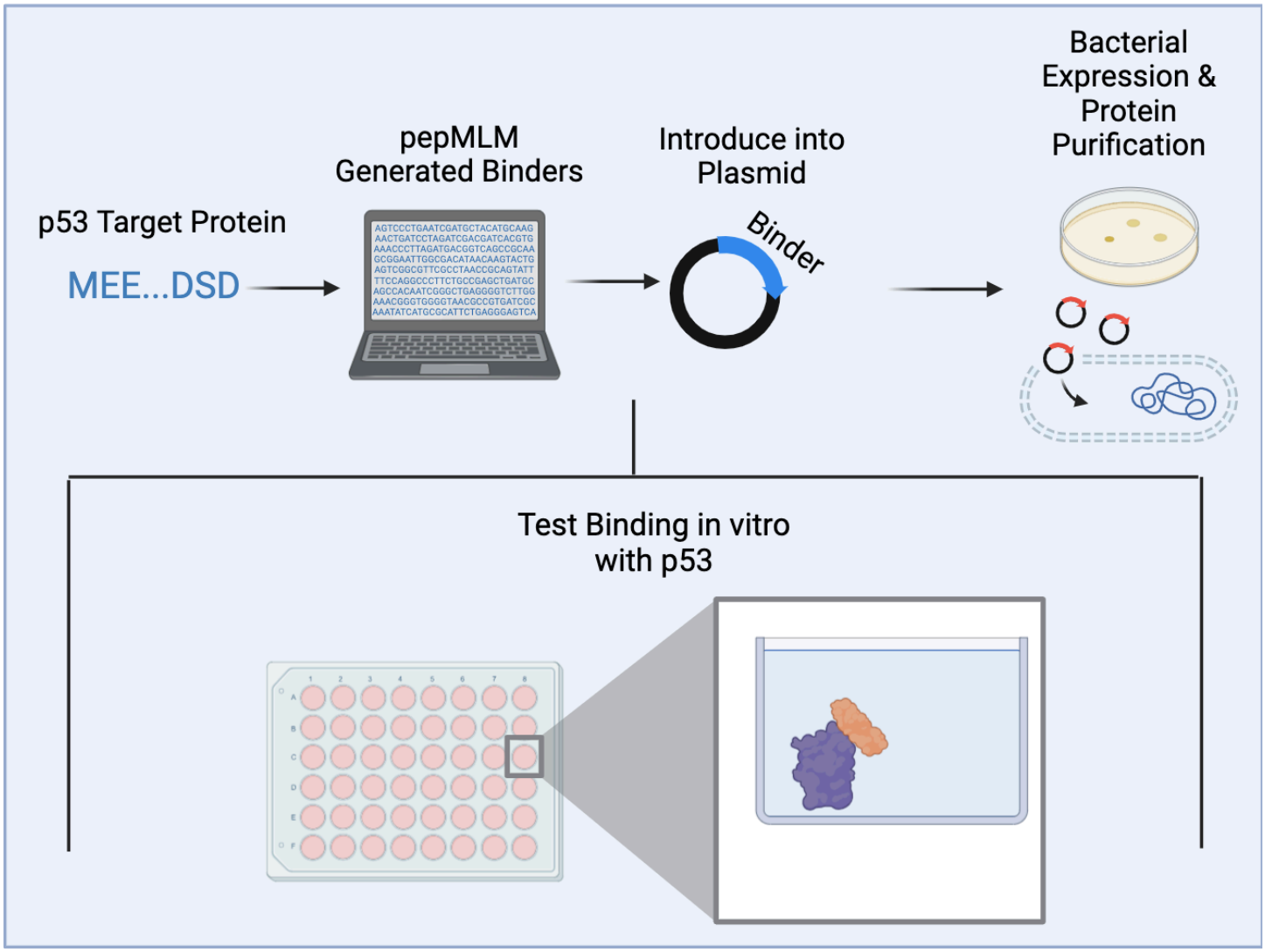


Figure 1. Graphical Abstract: workflow for computational design and experimental validation of p53-binding peptides. The amino acid sequence of p53 (starting with MEE...DSD) was input into the PepMLM algorithm to generate candidate peptide binders through sequence-based deep learning. Selected binder sequences were cloned into bacterial expression vectors containing necessary fusion tags (SUMO, His-tag) and linker sequences. The constructs were transformed into E. coli for protein expression, followed by affinity chromatography purification of the peptide binders. Binding affinity between the purified peptide and p53 was evaluated using enzyme-linked immunosorbent assay (ELISA), where varying concentrations of biotinylated peptide binder were tested against immobilized p53 protein to determine binding kinetics. This integrated pipeline combines computational design with experimental validation to develop peptide-based modulators of p53 function.

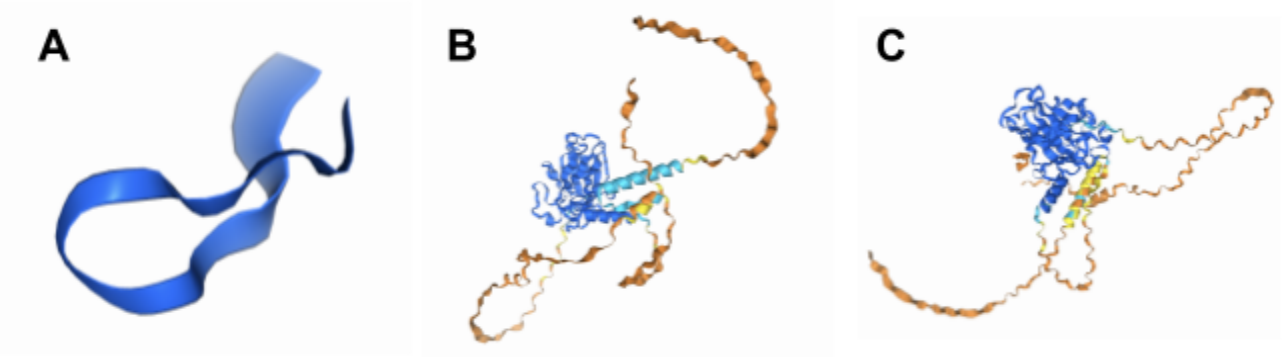


Figure 2. AlphaFold3 predicted structures of p53 binder,, and binder::p53 interaction. Shading was done according to AlphaFold's confidence metric, plDDT, as follows: Very low (plDDT < 50) = Orange, Low (70 > plDDT > 50) = Yellow, Confident (70 > plDDT > 90) = Light Blue, Very high (plDDT > 90) = Dark Blue. (A) Predicted structure of the candidate binder binder, with a pTM score of 0.02. (B) Predicted structure of p53, with a pTM score of 0.55. (C) Predicted binding interaction between p53 and the candidate binder. AlphaFold3 Predicts the binder interacts with p53 in the highly structured region of p53. However, the ipTM score of this interaction is low, at 0.25, which indicates AlphaFold3's low confidence in this interaction.

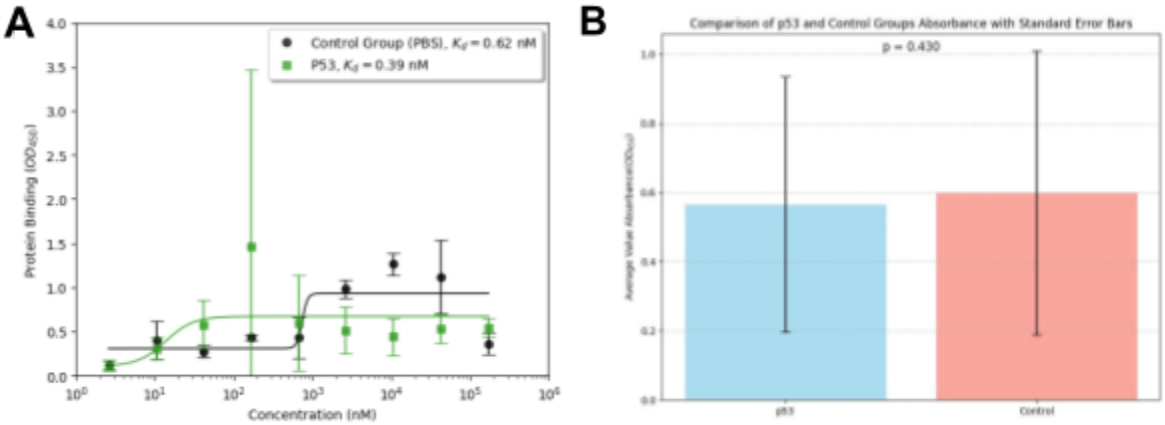


Figure 3. ELISA results comparing candidate binder affinity with P53 and candidate-PBS controls. (A) Data points represent mean absorbance values (OD_{450}) from three independent replicates at each concentration, with error bars showing standard deviation. Data fit with 4-parameter logistic regression to standard Kd curves[13]. ELISA's were performed with (n=3) replicates for each concentration, yielding a K_d (p53) = 0.3905 nM and a Kd (control) = 0.6198 nM. The solid lines represent the fitted curves for p53 (green) and control (black) conditions. (B) Statistical analysis of the obtained ELISA results, showing no statistically significant difference between the candidate binder's affinity for p53 and the negative control. A one-tailed paired t-test was performed manually (n=9), obtaining a t-statistic of -0.18 and p-value of 0.43.